Table of Contents

1. Introduction — What Are Control Laws?
2. Boeing 737-400 Command Speed Limiting
3. Boeing 737-400 Reversion Modes
4. Flight Envelope Protection (Fly-By-Wire)
5. High Angle of Attack Protection
6. High Speed Protection
7. Pitch Attitude Protection
8. Bank Angle Protection
9. Load Factor Protection
10. Autopilot Gain Adaption / Gain Scheduling
11. Practice Questions & Detailed Answers

GHOST AVIATOR

GHOST AVIATOR

Join live classes for guided practice and doubt clearing.

1. Introduction — What Are Control Laws?

Control Laws are the rules that govern autopilot authority — ensuring manoeuvres are proportional to the deviation, efficient, and within the safe flight envelope.

The autopilot must manoeuvre the aircraft logically and safely like a human pilot. This means:

- Not breaking aircraft limitations (speed, load factor, pitch and bank limits)
- Using enough of the aircraft's performance so manoeuvres are timely
- Scaling control response in proportion to the deviation/correction required — never commanding an excursion from the safe flight envelope

GHOST AVIATOR

Example: For an 80° turn, the autopilot would use 30°–45° bank. For a 10° turn, 30° bank would be too harsh. Control laws encode this proportional logic — small error → gentle correction; large error → stronger correction, but never beyond limits.

2. Boeing 737-400 Command Speed Limiting

What this section covers: How the AFS prevents flight envelope excursions through speed limiting.

The AFS provides speed, pitch and thrust commands to prevent exceeding the following limit speeds:

- VMO / MMO
- Wing flap limiting speeds
- Landing gear speeds
- Minimum speeds

GHOST AVIATOR

Rules:

- Commanded speed can equal but NOT exceed a limit speed
- Speeds greater than VMO/MMO cannot be selected from the MCP
- Speeds exceeding flap/gear limits *can* be selected — but AFS limits to the lower placard
- Minimum speed ≈ **1.3 VS** for current flap configuration (sensed by AoA vanes on forward fuselage)

If a speed greater than a placard speed or less than minimum speed is selected, the AFS accelerates/decelerates to *slightly short of the limit*, then commands the limit speed. An over-speed or under-speed limiting symbol appears in the MCP IAS/Mach display.

Join live classes for guided practice and doubt clearing.

To remove the under-speed symbol: select a speed **15 kt greater** than the minimum speed. A speed **5 kt greater** than minimum is what the AFS actually commands when limiting.

> **Special Case — V/S mode at minimum speed:** The A/P disengages and FD command bars retract when in LVL CHG climb with a command speed equal to minimum speed and a minimum rate of climb cannot be maintained without decelerating.

3. Boeing 737-400 Reversion Modes

> **What this section covers:** Automatic reversion to a more effective AFDS/A/T combination when speed control alone is insufficient.

GHOST AVIATOR

During some situations, speed control by AFDS or A/T alone could be insufficient. Mode reversion occurs *slightly before* reaching the limit speed. Two reversion modes exist:

Placard Limit Reversion

- NOT in AFDS/A/T speed control, A/T armed → A/T reverts to SPEED mode, controls speed to placard limit
- IN AFDS/A/T speed control → no reversion needed; system holds speed below placard
- A/T NOT available → no reversion for gear/flap limits; AFDS reverts to speed control for VMO/MMO only

Minimum Speed Reversion

- AFS commands speed **5 kt greater** than minimum
- If in V/S mode and speed drops to minimum → AFDS reverts to LVL CHG
- AFDS commands nose-down pitch to increase airspeed if thrust levers not advanced
- No minimum speed reversion when A/T is OFF and AFDS is in ALT HOLD, ALT ACQ, or after G/S capture

GHOST AVIATOR

4. Flight Envelope Protection (Fly-By-Wire)

> **Flight Envelope Protection** (FBW aircraft only) ensures the aircraft remains within the normal flight envelope in ALL phases of flight. The control laws prevent envelope violation during extreme situations.

Extreme situations protected against:

- Windshear
- Very high turbulence
- Mid-air collision avoidance (TCAS RA)
- GPWS or TCAS activation

Join live classes for guided practice and doubt clearing.

- Crew mismanagement

Protection purpose:

- Give full authority to allow best possible performance
- Reduce risk of overcontrolling / overstressing
- Provide easy, instinctive, immediate best-performance procedure

Parameters protected:

- Angle of attack
- Speed
- Pitch attitude
- Bank angle
- Load factor

GHOST AVIATOR

5. High Angle of Attack Protection

Enables rapid pull-up at maximum AoA (α_{max}) without overcontrolling. Technique: "snatch stick fully back." The pitch auto-trim stop alerts the pilot when exceeding the normal envelope. Auto-thrust automatically provides TOGA thrust at α_{floor} (before reaching α_{max}).

- **Input:** Angle of attack
- **Output:** Applied to elevators and auto-thrust

6. High Speed Protection

Prevents reaching Vd/Md by adding a positive nose-up g demand proportional to speed overshoot beyond VMO/MMO. A pilot can enter a steep dive knowing the system will prevent exceeding design speed limits.

- **Input:** Airspeed/Mach from ADC
- **Output:** Applied to elevators

GHOST AVIATOR

7. Pitch Attitude Protection

Only available in fly-by-wire aeroplanes. Enhances high AoA and high speed protection by reducing stick pitch demand when aircraft reaches pre-defined maximum pitch attitudes:

- Maximum pitch up: **30° nose-up**
- Maximum pitch down: **15° nose-down**
- **Input:** Pitch angle from attitude gyros
- **Output:** Applied to elevators

Join live classes for guided practice and doubt clearing.

8. Bank Angle Protection

Only available in fly-by-wire aeroplanes. Allows efficient roll manoeuvres while preventing uncontrollable states.

Condition	Bank Angle Limit
Normal flight envelope	**67°**
High AoA triggered	**45°**
High speed protection triggered	**40°**
Stick released after roll manoeuvre	Returns to **33°**

Bank angle limit is achieved by progressively reducing the roll rate demand as bank angle increases.

GHOST AVIATOR

9. Load Factor Protection

Available in fly-by-wire aeroplanes. Accelerometers sense g load. The limiter maintains aircraft within structural limitations while allowing immediate response to evasive manoeuvres. Linked to the high angle of attack protection.

- **Input:** Accelerometers (g load sensing)
- **Output:** Applied to elevators

10. Autopilot Gain Adaption / Gain Scheduling

> **What this section covers:** How the autopilot's authority is scaled based on flight regime, altitude and airspeed.

GHOST AVIATOR

Comparable to the Flight Director's gain scheduling (Chapter 25), the autopilot similarly reduces authority in critical phases:

Phase	Example Bank Angle Limit
Manual / cruise mode	45°
VOR / Localizer tracking	30°
Final approach / autoland	15°

Gain adaption also accounts for differing performance at altitude and speed. An input from the ADC (proportional to dynamic pressure Q) reduces autopilot authority at high speed to prevent overstressing — even though

Join live classes for guided practice and doubt clearing.

artificial feel protects the pilot, the autopilot could ignore feel and overstress the structure without this limiting.

Quick Revision Summary — Chapter 30:

- Control laws = proportional response rules; prevent flight envelope excursion
- Minimum speed ≈ 1.3 VS; AFS commands +5 kt above minimum; +15 kt removes under-speed symbol
- Reversion: Placard (gear/flap/VMO/MMO) and Minimum Speed (reverts to LVL CHG)
- FBW envelope protection: AoA, speed, pitch, bank, load factor
- High AoA: elevator + auto-thrust (α-floor = TOGA); technique = snatch stick back
- High Speed: nose-up g demand proportional to overshoot beyond VMO/MMO
- Pitch limits: 30° nose-up / 15° nose-down (FBW only)
- Bank limits: 67° normal / 45° high AoA / 40° high speed / returns to 33° on release (FBW)
- Load factor: accelerometer-based g limiting (FBW)
- Gain scheduling: reduces authority during autoland approach and at high Q (high speed)

GHOST AVIATOR

GHOST AVIATOR

Join live classes for guided practice and doubt clearing.

Practice Questions & Detailed Answers

Instructor-generated questions in DGCA CPL/ATPL examination style.

Q1. In a Boeing 737-400, the minimum speed protection activates at approximately:

a. 1.2 VS for current configuration

b. 1.3 VS for current configuration

c. VMC (minimum control speed)

d. VREF for current configuration

GHOST AVIATOR

Correct Answer: **(b)** 1.3 VS for current configuration

Explanation: The AFS minimum speed is based on angle of attack and is approximately **1.3 VS** for the current flap configuration, sensed by AoA vanes on either side of the forward fuselage. See Section 2.

Why other options are wrong:

- **(a)** 1.2 VS is the stall speed margin used in performance calculations, not the AFS minimum speed.
- **(c)** VMC is an engine-out control speed concept, not the AFS threshold.
- **(d)** VREF is approach speed, not the minimum speed protection threshold.

GHOST AVIATOR

Instructor's Note: *The AFS commands a speed 5 kt above the minimum, and the under-speed symbol disappears only when 15 kt above minimum is selected.*

Q2. In a fly-by-wire aircraft with bank angle protection, if the pilot releases the sidestick after a roll manoeuvre, the aircraft will return to a bank angle of:

a. 0° (wings level)

Join live classes for guided practice and doubt clearing.

b. 15°

c. 33°

d. 45°

Correct Answer: **(c)** 33°

Explanation: After a roll manoeuvre, if the pilot releases the stick, the aircraft returns to a bank angle of **33°**. The bank angle protection is not an "auto-wings-level" — it simply stops the bank from increasing beyond the limits. See Section 8.

Why other options are wrong:

- **(a)** A conventional autopilot might return to wings level, but FBW bank angle protection returns to 33°.
- **(b)** 15° is the autoland pitch limit, not the bank angle release position.
- **(d)** 45° is the high-AoA bank limit, not the release position.

Instructor's Note: *This is a key FBW distinction: stick neutral = maintain 33° bank, not wings level. "Normal law" FBW is load factor demand, not attitude demand.*

Q3. High speed protection on a fly-by-wire aircraft prevents exceeding Vd/Md by:

a. Automatically retarding the thrust levers

b. Deploying the speed brakes automatically

c. Adding a nose-up g demand proportional to overshoot beyond VMO/MMO

d. Limiting elevator deflection to nose-down only

Correct Answer: **(c)** Adding a nose-up g demand proportional to overshoot beyond VMO/MMO

Explanation: High speed protection adds a positive nose-up g demand proportional to the amount of speed overshoot beyond VMO/MMO. Input = airspeed/Mach from ADC; output to elevators. See

Join live classes for guided practice and doubt clearing.

Section 6.

Why other options are wrong:

- **(a)** Auto-thrust retard is a separate function; it is not the high speed protection mechanism.
- **(b)** Speed brakes are not automatically deployed by the high speed protection law.
- **(d)** Limiting to nose-down only would worsen overspeeds — the opposite of protection.

Instructor's Note: *The g demand is proportional — the further beyond VMO/MMO, the stronger the nose-up pitch input. This allows steep dives knowing the system will auto-recover.*

GHOST AVIATOR

Q4. Autopilot pitch and roll authority is significantly reduced as the aircraft nears the ground during autoland. This is an example of:

a. Fail-passive operation

b. Flight envelope protection

c. Gain adaption / gain scheduling

d. Minimum speed reversion

Correct Answer: **(c)** Gain adaption / gain scheduling

Explanation: Gain adaption / gain scheduling reduces the autopilot's pitch and roll authority during the final phase of automatic approach and autoland. The bank angle limit progressively reduces from ~45° in manual mode, to 30° for LOC/VOR tracking, to 15° for autoland finals. See Section 10.

Why other options are wrong:

- **(a)** Fail-passive refers to system failure behaviour (no unsafe deviation), not authority limits during normal operation.
- **(b)** Flight envelope protection is a FBW function; gain scheduling applies to conventional autopilots too.
- **(d)** Minimum speed reversion is an AFS mode change, not an authority limitation.

Join live classes for guided practice and doubt clearing.

> ***Instructor's Note:*** *Gain scheduling was introduced in Chapter 25 for FDS and applies equally to the autopilot — reducing authority near the ground and at high Q to prevent dangerous manoeuvres.*

Q5. On a fly-by-wire aircraft with pitch attitude protection, what are the maximum pitch attitude limits?

a. 25° nose-up and 10° nose-down

b. 30° nose-up and 15° nose-down

c. 20° nose-up and 20° nose-down

d. 45° nose-up and 30° nose-down

GHOST AVIATOR

Correct Answer: **(b)** 30° nose-up and 15° nose-down

Explanation: Pitch attitude protection limits are **30° nose-up** and **15° nose-down**. See Section 7.

Why other options are wrong:

- **(a), (c), (d)** These are fabricated values. The specific limits of 30° nose-up and 15° nose-down must be memorised.

> ***Instructor's Note:*** *Asymmetric limits — nose-up is more than nose-down — reflect that overpitch in the dive (nose-down) is corrected more aggressively (hits speed limits faster) than nose-up, which has AoA protection as a further back-stop.*

GHOST AVIATOR

Q6. During a 737-400 AFS minimum speed reversion while in V/S (Vertical Speed) mode, the AFDS will automatically revert to:

a. ALT HOLD

b. LVL CHG

c. Speed (SPD) mode

Join live classes for guided practice and doubt clearing.

d. ARM mode

Correct Answer: **(b)** LVL CHG

Explanation: If operating in V/S mode and actual speed becomes equal to or slightly less than the minimum speed, the under-speed limiting symbol appears in the MCP IAS/Mach display and the AFDS reverts to **LVL CHG**. The AFDS then commands nose-down pitch to increase airspeed if thrust levers are not advanced. See Section 3.

Why other options are wrong:

- **(a)** ALT HOLD is an altitude mode and would not increase airspeed in a deceleration scenario.
- **(c)** Speed mode is an A/T mode, not an AFDS pitch mode reversion for minimum speed.
- **(d)** ARM is the A/T state when no A/T mode is engaged — not an AFDS pitch reversion mode.

Instructor's Note: *No minimum speed reversion is available when A/T is OFF and AFDS is in ALT HOLD, ALT ACQ, or after G/S capture — this is an important limitation to note.*

GHOST AVIATOR

Join live classes for guided practice and doubt clearing.